pipes because they have, you guessed it, chambers instead of pipes for the water to evaporate and travel from a heated up area to a colder region and condense, releasing the heat externally. While smartphone manufacturers have stayed away from vapour chambers due to the additional cost and the bulk they would add to a smartphone, recent advancements have made them both cheap and sleek enough to not put much additional bulk into the design. Expect to see them hit mainstream phones soon enough, as the cooling with vapour chambers is more effective than the heatpipe counterpart.

Polymer film

Employing the electrocaloric effect that uses electric fields to change the temperature of a material, scientists at UCLA and SRI International have devised a thin, flexible polymer material that can be used to reduce the temperature of devices significantly. The polymer transports the heat from the source to the sink when needed by alternating contacts between them. Think of it like this – when a piece of metal is heated on one end, the constituent particles on that end are in a higher state of excitement than the particles on the other end. An electric field that makes these particles flow from the hot end to the cold end would inevitably spread out the heat evenly and reduce the temperature further. A simple way to visualise this is how, after you've added some cold water to a bucket of hot water, mixing it manually with your hand makes it cool faster. This is the principle that makes this technology work. Due to the flexibility of the polymer field, this device will work with smartphones that are curved and flexible as well.

PCs

This one is a little more obvious to us geeks. If you're a veteran of the master race, who has wrangled with the flaming beasts that lie within your cabinet during a session of the latest AAA game or a 4K video render, you know important it is to keep a PC cool. And the future might have some interesting developments in cooling tech!

Cooling centric design

One of the first places where manufacturers look to improve the cooling capabilities of PCs is the build itself. And when we say build, we are referring to the material of choice, the cabinet design and more. Take, for instance, the Streacom DB6 that was revealed at Computex in 2017. The fanless tower case is a microATX (340 x 360 x 245mm) chassis that will be able to passively cool a 125W CPU and a 125W GPU. We say 'will' as the product hasn't been listed commercially yet. The design features side panels made of aluminium that push the weight of the case to a whopping 18kgs but it does an exceptional job of keeping the system cool. The loop heat pipe system on the interior, also made of metal, is technology that is used in satellites and industrial applications and hasn't yet been brought to mainstream PC applications. Although, similar aluminium builds are also available from other manufacturers.

Streacom DB6 with its aluminium side panels

Complete immersion

It is one thing to liquid cool your system the usual way, and a whole different ball game to immerse it completely into a liquid. Sounds crazy? It actually isn't, unless you plan on using water as the liquid of choice. In a two-phase immersion cooled system, the media of choice for immersion are generally dielectric heat transfer liquids, which are much better heat conductors than air, water or oil. The electronic components are submerged into a bath of the same. With their various low boiling points (ie, 49°C vs. 100°C in water), the fluids rapidly boil on the surface of heat generating components and rising vapor passively takes care of heat transfer. One example of such a liquid is the 3M Novec 649 Engineered Fluid. It offers some direct advantages over other such media by being environment friendly and clean – oil isn't exactly ideal if you would like to keep your system neat.

Source: CNET

An immersion cooling setup

Advanced liquid cooling

There are a few ways in which the usual liquid cooling methods can be taken up a notch. Phase-change cooling essentially converts your PC into a refrigerator. A vapor compression

NSG S0 co-designed with Watermod, a complete-passive chassis designed to cool components without the need of fans.